

Day 24: Data Refresh & On-Premises Gateways

📌 Definition to Remember

The **On-Premises Data Gateway** acts as a secure cryptographic bridge that enables Power BI Service cloud datasets to query and refresh data from local on-premises databases and network file shares.

1. Gateway Types & Architecture

- **Standard Mode (Enterprise Gateway):** Installed on a dedicated server; supports multiple users, scheduled refreshes, and DirectQuery live connections across enterprise SQL and Oracle servers.
- **Personal Mode:** Installed on a single developer's workstation; supports only scheduled refresh for one user.

```
[Power BI Service Cloud]
  | (Encrypted HTTPS Port 443)
  ▼
[On-Premises Gateway Service] (Behind Corporate Firewall)
  | (Local Network Query)
  ▼
[Local SQL Server / Excel Files]
```

2. Scheduled Refresh Limits

- **Power BI Pro:** Up to **8 scheduled refreshes per day** per semantic model.
- **Power BI Premium / Fabric:** Up to **48 scheduled refreshes per day** (or sub-minute with incremental refresh).

3. Incremental Refresh

- Instead of reloading 10 years of historical data on every daily refresh, Incremental Refresh:
- Archives historical partitions (read-only).
- Only queries and refreshes the last N days of modified data (**RangeStart** and **RangeEnd** parameters).

★ Must-Master Skills & Interview Points

1. How the On-Premises Gateway works without opening inbound corporate firewall ports.
2. Diagnosing common refresh failures (expired credentials, network timeouts, M query folding breaks).
3. Configuring Incremental Refresh with **RangeStart** and **RangeEnd**.

⚡ Quick Recall

Enterprise Gateway (Secure Firewall Bridge) + Scheduled Refresh (8x Pro / 48x Premium) + Incremental Refresh

